Firms, Production Possibility Sets, and Profit Maximization

Econ 2100, Fall 2019

Lecture 1, 21 October

Outline

1. Logistics
2. Production Sets and Production Functions
3. Profits Maximization, Supply Correspondence, and Profit Function

Logistics

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Class Time and Location
Monday & Wednesday, 10:30am to 11:45am in WWPH 4940
Recitation Times
Friday, 2pm to 3:15am, in WWPH 4940.

Class Webpage `http://www.pitt.edu/~luca/ECON2100/`

Announcements
None Today

Logistics

Handouts: *For each class, I will post an handout that you can use to take notes (either print and bring to class, or use computer).*

Textbooks: *Kreps, Microeconomic Foundations I: Choice and Competitive Markets, Princeton University Press*
Mas-Collel, Whinston and Green: Microeconomic Theory.
Other useful textbooks by Varian, Jehle and Reny, older Kreps.
My favorite: Gerard Debreu: Theory of Value.

Grading

- *Luca's Half: Problem set 10%, Exam 90%*
  Final Exam, 2 hours long, date TBA

- Problem Sets
  - Weekly, some questions appear during class.
  - **DO NOT** look for answers... These are for you to learn what you can and cannot do; we can help you get better only if we know your limits.
  - Working in teams strongly encouraged, but turn in problem sets individually.
  - Start working on exercises on your own, and then get together to discuss.
  - Typically, a Problem Set will be due at the beginning of each Monday class.

Luca's Goals for Micro Theory Sequence

- Learn and understand the microeconomic theory every academic economist needs to know.
- Stimulate interest in micro theory as a field (some of you may want to become theorists).
- Enable you to read papers that use theory, and go to research seminars.
- We will cover the following standard topics:
  - firms: production, profit maximization, aggregation;
  - general equilibrium theory: consumers and firms in competitive markets, Walrasian equilibrium, First and Second Welfare Theorems, existence of equilibrium, uniqueness of equilibrium, markets with uncertainty and time, Arrow-Debreu economies.
- In the Spring, Professor Richard van Weelden will cover game theory and information economics.
- Questions? This will be fun: let's go!

Producers and Production Sets

Producers are profit maximizing firms that buy inputs and use them to produce and then sell outputs.

- The plural is important because most firms produce more than one good.

- The standard undergraduate textbook description focuses on one ouput and a few inputs (two in most cases).
- In that framework, production is described by a function that takes inputs as the domain and output(s) as the range.
- Here we focus on a more general and abstract version of production.

Definition

A production set is a subset $Y \subseteq \mathbf{R}^n$.

- $y = (y_1, ..., y_N)$ denotes production (input-output) vectors.
- A production vector has outputs as non-negative numbers and inputs as non-positive numbers: $y_i \leq 0$ when i is an input, and $y_i \geq 0$ if i is an output.
- What is $p \cdot y$ in this notation?

Production Set Properties

Definition

$Y \subseteq \mathbf{R}^n$ satisfies:

- no free lunch if $Y \cap \mathbf{R}^n_+ \subseteq \{\mathbf{0}_n\}$;
- possibility of inaction if $\mathbf{0}_n \in Y$;
- free disposal if $y \in Y$ implies $y' \in Y$ for all $y' \leq y$;
- irreversibility if $y \in Y$ and $y \neq \mathbf{0}_n$ imply $-y \notin Y$;
- nonincreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \in [0, 1]$;
- nondecreasing returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 1$;
- constant returns to scale if $y \in Y$ implies $\alpha y \in Y$ for all $\alpha \geq 0$;
- additivity if $y, y' \in Y$ imply $y + y' \in Y$;
- convexity if Y is convex;
- Y is a convex cone if for any $y, y' \in Y$ and $\alpha, \beta \geq 0$, $\alpha y + \beta y' \in Y$.

Draw Pictures.

Production Set Properties Are Related

- Some of these properties are related to each other.

Exercise

Y satisfies additivity and nonincreasing returns if and only if it is a convex cone.

Exercise

For any convex $Y \subset \mathbf{R}^n$ such that $\mathbf{0}_n \in Y$, there is a convex $Y' \subset \mathbf{R}^{n+1}$ that satisfies constant returns such that $Y = \{y \in \mathbf{R}^n : (y, -1) \in \mathbf{R}^{n+1}\}$.

Production Functions

Let $y \in \mathbf{R}_+^m$ denote outputs while $x \in \mathbf{R}_+^n$ represent inputs; if the two are related by a function $f : \mathbf{R}_+^n \to \mathbf{R}_+^m$, we write $y = f(x)$ to say that y units of outputs are produced using x amount of the inputs.

- When $m = 1$, this is the familiar one output many inputs production function.
- Production sets and the familiar production function are related.

Exercise

Suppose the firm's production set is generated by a production function $f : \mathbf{R}_+^n \to \mathbf{R}_+^m$, where $\mathbf{R}_+^n$ represents its n inputs and $\mathbf{R}_+$ represents its m outputs. Let

$$Y = \{(-x, y) \in \mathbf{R}_-^n \times \mathbf{R}_+^m : y \leq f(x)\}.$$

Prove the following:

1. Y satisfies no free lunch, possibility of inaction, free disposal, and irreversibility.
2. Suppose $m = 1$. Y satisfies constant returns to scale if and only if f is homogeneous of degree one, i.e. $f(\alpha x) = \alpha f(x)$ for all $\alpha \geq 0$.
3. Suppose $m = 1$. Y satisfies convexity if and only if f is concave.

Transformation Function

- We can describe production sets using a function.

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the transformation function $F : Y \to \mathbf{R}$ is defined by

$$Y = \{y \in Y : F(y) \leq 0 \text{ and } F(y) = 0 \text{ if and only if } y \text{ is on the boundary of } Y\};$$

the transformation frontier is

$$\{y \in \mathbf{R}^n : F(y) = 0\}$$

Definition

Given a differentiable transformation function F and a point on its transformation frontier y, the marginal rate of transformation for goods i and j is given by

$$MRT_{i,j} = \frac{\frac{\partial F(y)}{\partial y_i}}{\frac{\partial F(y)}{\partial y_j}}$$

- Since $F(y) = 0$ we have $$\frac{\partial F(y)}{\partial y_i} dy_i + \frac{\partial F(y)}{\partial y_j} dy_j = 0$$
  - MRT is the slope of the transformation frontier at y.

Profit Maximization

Profit Maximizing Assumption

The firm's objective is to choose a production vector on the trasformation frontier as to maximize profits given prices $p \in \mathbf{R}^n_{++}$:

$$\max_{y \in Y} p \cdot y$$

or equivalently

$$\max p \cdot y \qquad \text{subject to} \qquad F(y) \leq 0$$

- Does this distinguish between revenues and costs? How?
- Using the single output production function:

$$\max_{x \geq 0} pf(x) - w \cdot x$$

  - here $p \in \mathbf{R}_{++}$ is the price of output and $w \in \mathbf{R}^l_{++}$ is the price of inputs.

First Order Conditions For Profit Maximization

$$\max p \cdot y \qquad \textit{subject to} \qquad F(y) = 0$$

Profit Maximizing

- The FOC are

$$p_i = \lambda \frac{\partial F(y)}{\partial y_i} \text{ for each } i \qquad \text{or} \qquad \underbrace{p}_{1\times n} = \lambda \underbrace{\nabla F(y)}_{1\times n} \text{ in matrix form}$$

- Therefore

$$\frac{1}{\lambda} = \frac{\frac{\partial F(y)}{\partial y_i}}{p_i} \text{ for each } i$$

  - the marginal product per dollar spent or received is equal across all goods.

- Using this formula for i and j:

$$\frac{\frac{\partial F(y)}{\partial y_i}}{\frac{\partial F(y)}{\partial y_j}} = MRT_{i,j} = \frac{p_i}{p_j} \text{ for each } i, j$$

  - the Marginal Rate of Transformation equals the price ratio.

Supply Correspondence and Profit Functions

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the supply correspondence $y^* : \mathbf{R}^n_{++} \to \mathbf{R}^n$ is:

$$y^*(p) = \arg\max_{y \in Y} p \cdot y.$$

- Tracks the optimal choice as prices change (similar to Walrasian demand).

Definition

Given a production set $Y \subseteq \mathbf{R}^n$, the profit function $\pi : \mathbf{R}^n_{++} \to \mathbf{R}$ is:

$$\pi(p) = \max_{y \in Y} p \cdot y.$$

- Tracks maximized profits as prices change (similar to indirect utility function).

Proposition

If Y satisfies non decreasing returns to scale either $\pi(p) \leq 0$ or $\pi(p) = +\infty$.

Proof.

Question 6, Problem Set 5. □

Properties of Supply and Profit Functions

Proposition

Suppose Y is closed and satisfies free disposal. Then:

- $\pi(\alpha p) = \alpha\pi(p)$ *for all* $\alpha > 0$;
- π *is convex in* p;
- $y^*(\alpha p) = y^*(p)$ *for all* $\alpha > 0$;
- *if* Y *is convex, then* $y^*(p)$ *is convex;*
- *if* $|y^*(p)| = 1$, *then* π *is differentiable at* p *and* $\nabla\pi(p) = y^*(p)$ *(Hotelling's Lemma).*
- *if* $y^*(p)$ *is differentiable at* p, *then* $Dy^*(p) = D^2\pi(p)$ *is symmetric and positive semidefinite with* $Dy^*(p)p = 0$.

The Profit Function Is Convex

Proof.

Let $p, p' \in \mathbf{R}^n_{++}$ and let the corresponding profit maximizing solutions be y and y'.

- For any $\lambda \in (0,1)$ let $\bar{p} = \lambda p + (1-\lambda) p'$ and let $\bar{y}$ be the profit maximizing output vector when prices are $\bar{p}$.
- By "revealed preferences"

$$p \cdot y \geq p \cdot \bar{y} \qquad \text{and} \qquad p' \cdot y' \geq p' \cdot \bar{y}$$

  - why?
- multiply these inequalities by λ and $1-\lambda$

$$\lambda p \cdot y \geq \lambda p \cdot \bar{y} \qquad \text{and} \qquad (1-\lambda) p' \cdot y' \geq (1-\lambda) p' \cdot \bar{y}$$

- summing up

$$\lambda p \cdot y + (1-\lambda) p' \cdot y' \geq [\lambda p + (1-\lambda) p'] \cdot \bar{y}$$

- using the definition of profit function:

$$\lambda \pi(p) + (1-\lambda) \pi(p') \geq \pi(\lambda p + (1-\lambda) p')$$

proving convexity of $\pi(p)$. □

The Supply Correspondence Is Convex

Proof.

Let $p \in \mathbf{R}^n_{++}$ and let $y, y' \in y^*(p)$.
We need to show that if Y is convex then

$$\lambda y + (1-\lambda)\, y' \in y^*(p) \qquad \text{for any } \lambda \in (0,1)$$

- By definition:

$$p \cdot y \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y \qquad \text{and} \qquad p \cdot y' \geq p \cdot \bar{y} \quad \text{for any } \bar{y} \in Y$$

- multiplying by λ and $1-\lambda$ we get

$$\lambda p \cdot y \geq \lambda p \cdot \bar{y} \qquad \text{and} \qquad (1-\lambda)\, p \cdot y' \geq (1-\lambda)\, p \cdot \bar{y}$$

- Therefore, summing up, we have

$$\lambda p \cdot y + (1-\lambda)\, p \cdot y' \geq [\lambda + (1-\lambda)]\, p \cdot \bar{y}$$

- Rearranging:

$$p \cdot [\lambda y + (1-\lambda)\, y'] \geq p \cdot \bar{y}$$

proving convexity of $y^*(p)$. □

Hotelling's Lemma

if $|y^*(p)| = 1$, then π is differentiable at p and $\nabla\pi(p) = y^*(p)$

Proof.

Suppose $y^*(p)$ is the unique solution to $\max p \cdot y$ subject to $F(y) = 0$.

- The Envelope Theorem says

$$D_q\phi(x^*(\bar{q}); \bar{q}) = D_q\phi(x, q)|_{x=x^*(q), q=\bar{q}} - [\lambda^*(\bar{q})]^\top D_q F(x, q)|_{x=x^*(\bar{q}), q=\bar{q}}$$

- In our setting,
  - $\phi(x, q) = p \cdot y$,
  - $F(x, q) = F(y)$, and
  - $\phi(x^*(\bar{q}); \bar{q}) = \pi(p)$.
- Thus, by the envelope theorem:

$$\nabla\pi(p) = D_p(p \cdot y)|_{y=y^*(p)} - [\lambda^*(p)]^\top D_p F(y)|_{y=y^*(p)} = y|_{y=y^*(p)} - [\lambda^*(p)]^\top 0$$

  - because $p \cdot y$ is linear in p and $D_p F(y) = 0$.
- Therefore

$$\nabla\pi(p) = y^*(p)$$

as desired. □

Law of Supply

Remark

If $y^*(p)$ is differentiable at p, then $Dy^*(p) = D^2\pi(p)$ is positive semidefinite.

Write the Lagrangian

$$L = p \cdot y - \lambda F(y)$$

- *By the Envelope Theorem:*

$$\frac{\partial \pi(p)}{\partial p_i} = \left.\frac{\partial L}{\partial p_i}\right|_{y=y^*} = y_i^*(p)$$

- *Therefore, we have*

$$\frac{\partial^2 \pi(p)}{\partial p_i} = \frac{\partial y_i^*(p)}{\partial p_i} \geq 0$$

  - *where the inequality follows from convexity of the profit function.*

- This is called the Law of Supply: quantity responds in the same direction as prices.
  - Notice that here y_i can be either input or output.
    - What does this mean for outputs?
    - What does this mean for inputs?

Factor Demand, Supply, and Profit Function

- The previous concepts can be stated using the production function notation.

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's factor demand is

$$x^*(p, w) = \arg\max_x \{py - w \cdot x \text{ subject to } f(x) = y\} = \arg\max_x pf(x) - w \cdot x.$$

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's supply $y^* : \mathbf{R}^n_+ \to \mathbf{R}$ is defined by

$$y^*(p, w) = f(x^*(p, w)).$$

Definition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's profit function $\pi : \mathbf{R}_{++} \times \mathbf{R}^n_{++} \to \mathbf{R}$ is defined by

$$\pi(p, w) = py^*(p, w) - w \cdot x^*(p, w).$$

Factor Demand Properties

- Given these definitions, the following results "translate" the results for output sets to production functions.

Proposition

Given $p \in \mathbf{R}_{++}$ and $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$,

1. *$\pi(p, w)$ is convex in (p, w).*
2. *$y^*(p, w)$ is non decreasing in p (i.e. $\frac{\partial y^*(p,w)}{\partial p} \geq 0$) and $x^*(p, w)$ is non increasing in w (i.e. $\frac{\partial x_i^*(p,w)}{\partial w_i} \leq 0$) (Hotelling's Lemma).*

Proof.

Problem 7a,b; Problem Set 4. □

Cost Minimization

Cost Minimizing

- Consider the single output case and suppose the firm wants to deliver a given output quantity at the lowest possible costs. The firm solves

$$\min w \cdot x \qquad \textit{subject to} \qquad f(x) = y$$

- This has no simple equivalent in the output vector notation.

Definition

Given $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's conditional factor demand is

$$x^*(w, y) = \arg\min \{w \cdot x \text{ subject to } f(x) = y\};$$

Definition

Given $w \in \mathbf{R}^n_{++}$ and a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, the firm's cost function $C : \mathbf{R}^n_{++} \times \mathbf{R}_+ \to \mathbf{R}$ is defined by

$$C(w, y) = w \cdot x^*(w, y).$$

Properties of Cost Functions

Proposition

Given a production function $f : \mathbf{R}^n_+ \to \mathbf{R}_+$, *the corresponding cost function* $C(w, y)$ *is concave in* w.

Proof.

Question 7c; Problem Set 4. (Hint: use a 'revealed preferences' argument) □

Shephard's Lemma

- Write the Lagrangian

$$L = w \cdot x \ - \lambda \left[f(x) - y \right]$$

- by the Envelope Theorem

$$\frac{\partial C(w, y)}{\partial w_i} = \frac{\partial L}{\partial w_i} = x_i^* (w, y)$$

Conditional factor demands are downward sloping

Differentiating one more time: $\frac{\partial C^2(w,y)}{\partial w_i \partial w_i} = \frac{\partial x_i^*(w,y)}{\partial w_i} \leq 0$ where the inequality follows concavity of $C(w, y)$.

Next Class

- General equilibrium introduction and notation
- Pareto optimality